



NANYANG
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CZ3005 Artificial Intelligence

Week 12a – Fuzzy Logic

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Learning Goals

Understanding the:

- Basic definitions and terminology
- Set-theoretic operations
- Membership Function (MF) formulation
 - MFs parameterization
 - Linguistic modifier/hedges

Example: Safe Autonomous Vehicles

- Autonomous Cars implement Duty of Care
 - an individual should exercise “reasonable care” while performing acts that could harm others
- “On a Formal Model of Safe and Scalable Self-driving Cars”, by Shalev-Swartz, Shammah, and Shashua, arXiv 1708.06374
 - Responsibility Sensitive Safety – mathematical safety assurance
 - System design that adheres to the mathematical model



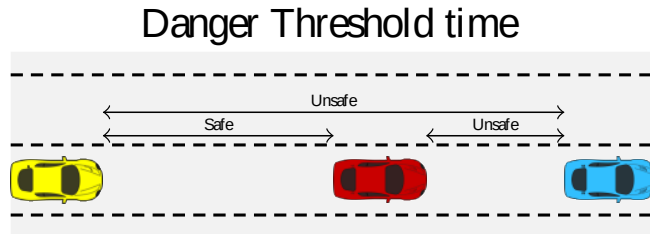
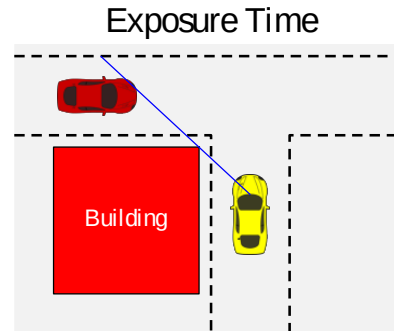
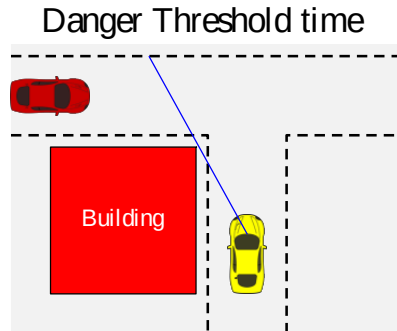
Example: Safe Autonomous Vehicles

- Responsibility Sensitive Safety (RSS)
 - Do not hit someone from behind
 - Do not cut-in recklessly
 - Right-of-way is given, not taken
 - Be careful of areas with limited visibility
 - If you can avoid an accident without causing another one, you must do so

Example: Safe Autonomous Vehicles

- Responsibility Sensitive Safety (RSS)
 - Do not hit someone from behind
 - Even if not your fault?
 - Do not cut-in recklessly
 - Right-of-way is given, not taken
 - How to resolve polite deadlocks?
 - Be careful of areas with limited visibility
 - If you can avoid an accident without causing another one, you must do so
 - Emergency breaking can cause whiplash

Example: Safe Autonomous Vehicles



“On a Formal Model of Safe and Scalable Self-driving Cars”,
by Shalev-Swartz, Shammah, and Shashua, arXiv 1708.06374

Example: Safe Autonomous Vehicles

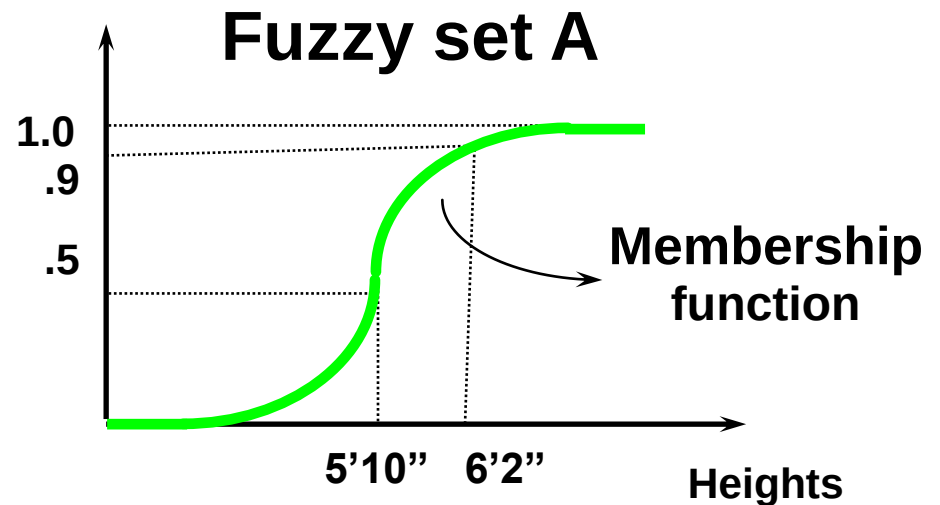
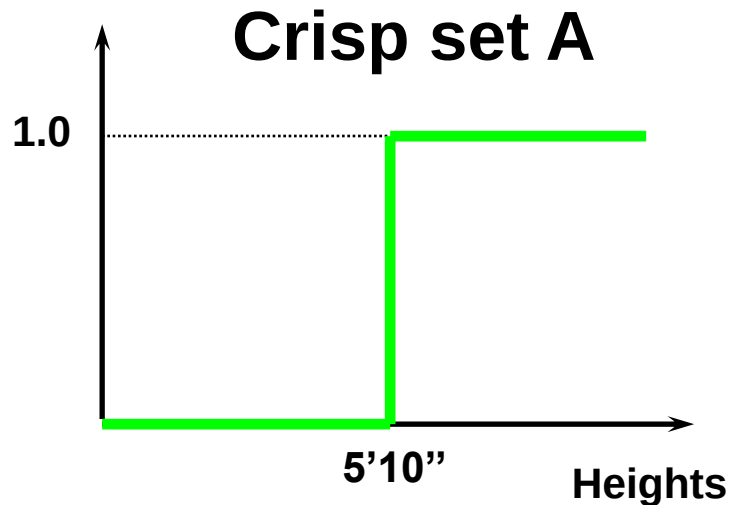
- Use of Semantic Action Space
 - Not “drive for 5.33 kilometers, then reduce speed at the rate of 1 m/s^2 ”
 - Slow down as you approach red light to stop at the line.
 - IF approach red light, THEN slow down and stop

"On a Formal Model of Safe and Scalable Self-driving Cars",
by Shalev-Swartz, Shammah, and Shashua, arXiv 1708.06374

Fuzzy Sets

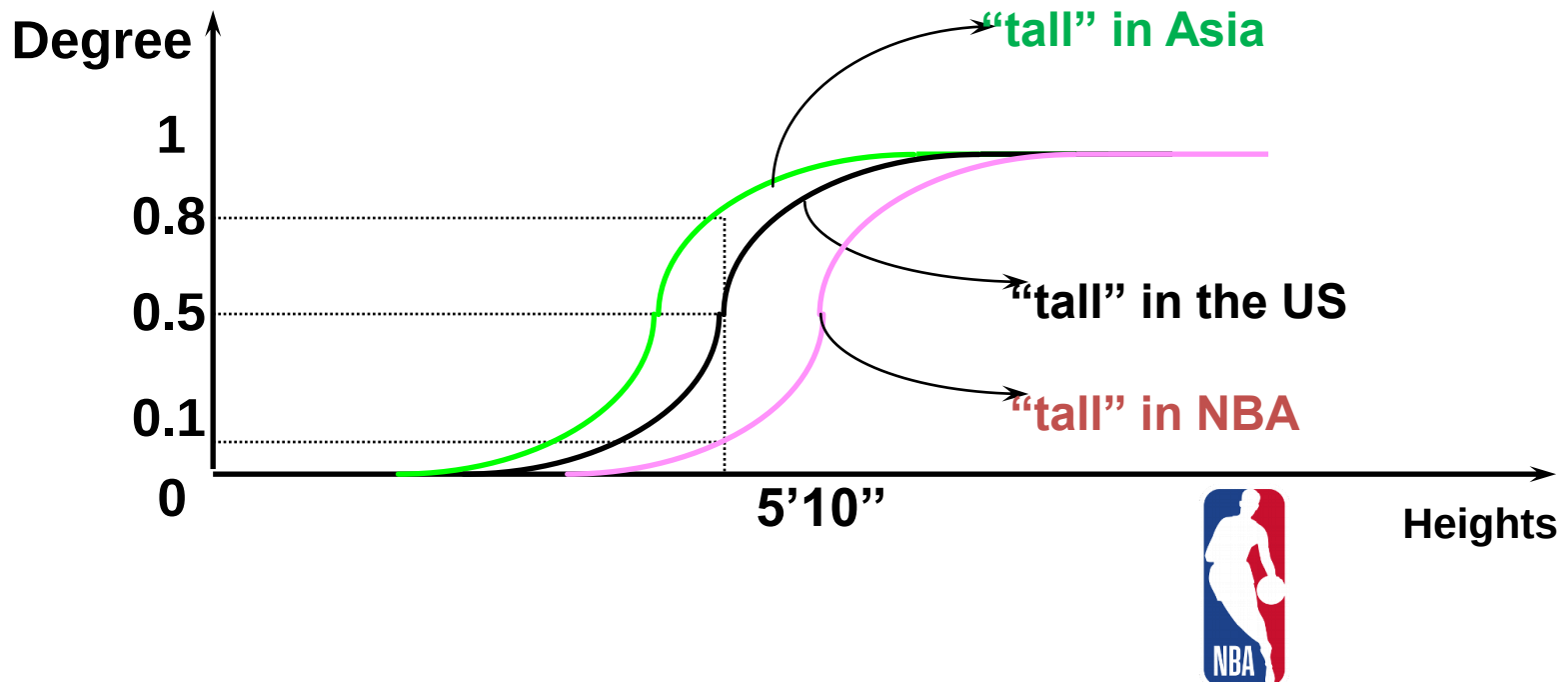
- Sets with fuzzy boundaries

A = Set of tall people



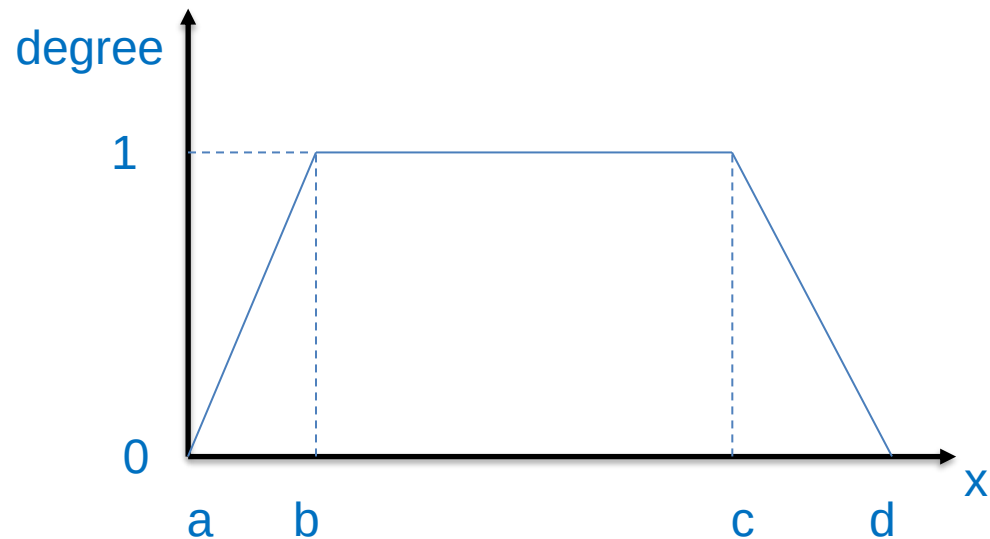
Fuzzy Membership Function

- Characteristics of a fuzzy MF:
 - Subjective measures between 0 and 1
 - Not probability functions



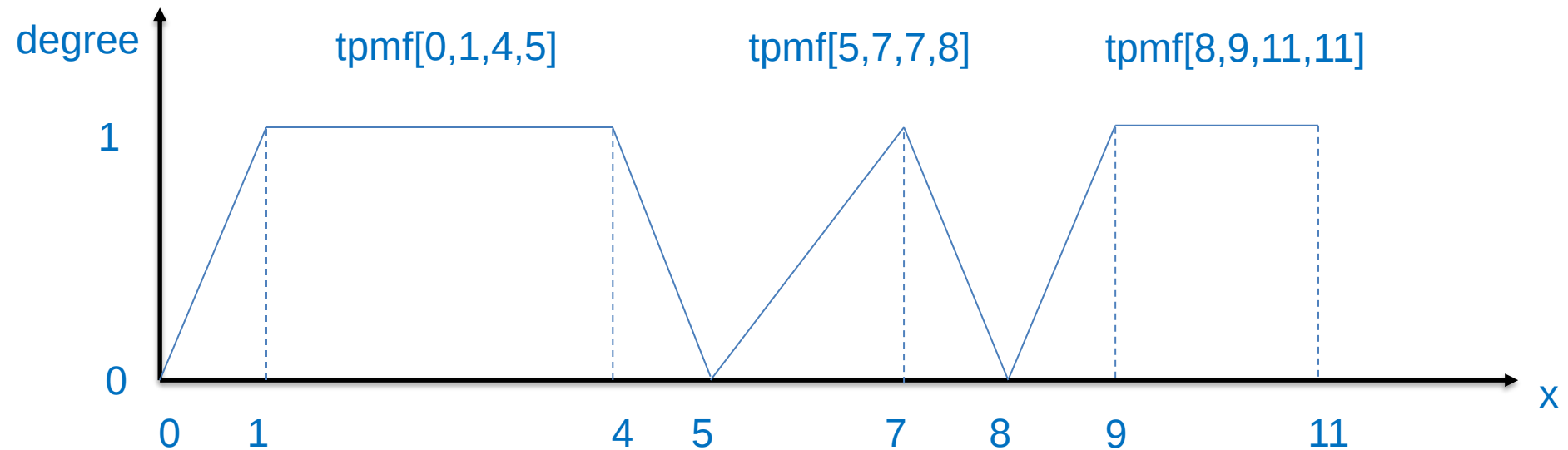
Fuzzy Membership Function

- Trapezoidal MF:
 - $tpmf[a,b,c,d]$
 - where
 - degree=0 at $x=a$
 - degree=1 at $x=b$
 - degree=1 at $x=c$
 - degree=0 at $x=d$



Fuzzy Membership Function

- Trapezoidal MF:
 - examples



Fuzzy Sets

Formal definition:

A fuzzy set A in X is expressed as a set of ordered pairs:

$$A = \{ \overbrace{(x, \underbrace{\mu_A(x)}_{\text{Membership function (MF)}})}^{\text{Crisp input}} \mid \underbrace{x \in X}_{\text{Universe or universe of discourse}} \}$$

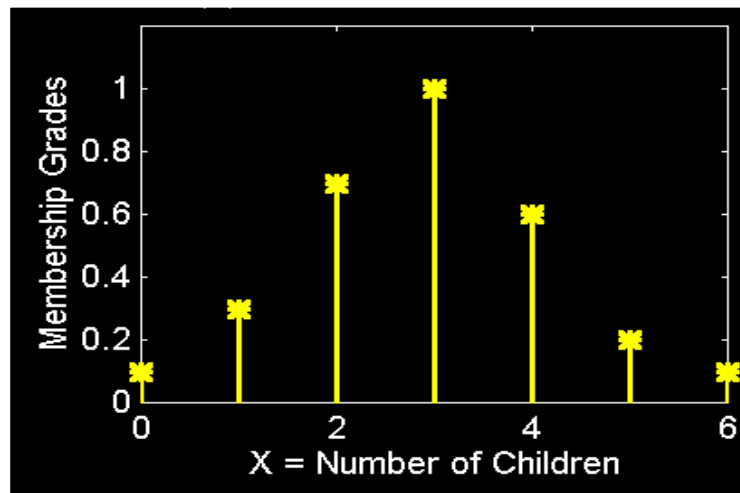
Diagram illustrating the formal definition of a fuzzy set A in X :

- Fuzzy set** points to the set notation $\{ \dots \}$.
- Crisp input** points to the element x .
- Membership function (MF)** points to the function $\mu_A(x)$.
- Universe or universe of discourse** points to the condition $x \in X$.

A fuzzy set is totally characterized by a membership function (MF).

Fuzzy Sets – Discrete Universes

- Fuzzy set C = “desirable city to live in”
 $X = \{\text{SF}, \text{Boston}, \text{LA}\}$ (discrete and non-ordered)
 $C = \{(\text{SF}, 0.9), (\text{Boston}, 0.8), (\text{LA}, 0.6)\}$
- Fuzzy set A = “sensible number of children to have”
 $X = \{0, 1, 2, 3, 4, 5, 6\}$ (discrete universe)
 $A = \{(0, .1), (1, .3), (2, .7), (3, 1), (4, .6), (5, .2), (6, .1)\}$



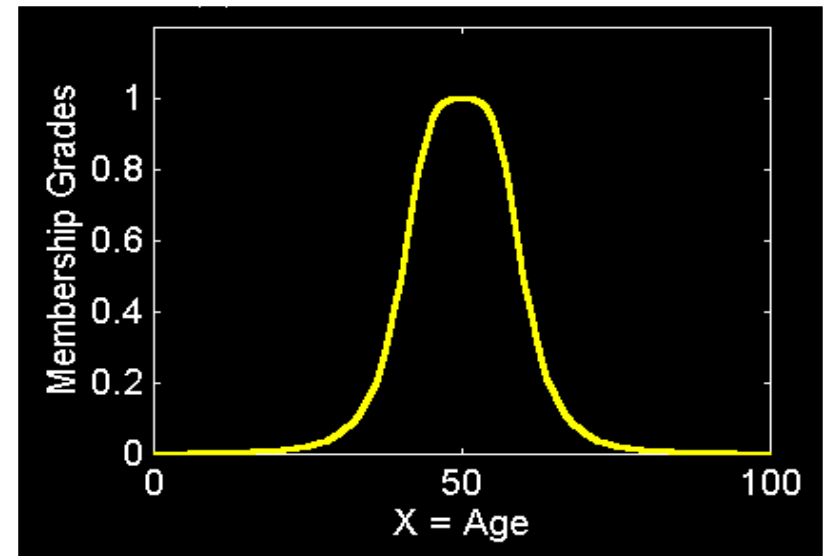
Fuzzy Sets – Continuous Universes

- Fuzzy set B = “about 50 years old”

X = Set of positive real numbers (continuous)

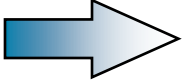
$$B = \{(x, \mu_B(x)) \mid x \text{ in } X\}$$

$$\mu_B(x) = \frac{1}{1 + \left(\frac{x - 50}{10}\right)^2}$$



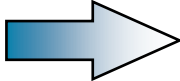
Alternative Notation

A fuzzy set A can be alternatively denoted as follows:

X is discrete 
$$A = \sum_{x_i \in X} \mu_A(x_i) / x_i$$

$X = \{0, 1, 2, 3, 4, 5, 6\}$ (discrete universe)

$A = \{(0, .1), (1, .3), (2, .7), (3, 1), (4, .6), (5, .2), (6, .1)\}$

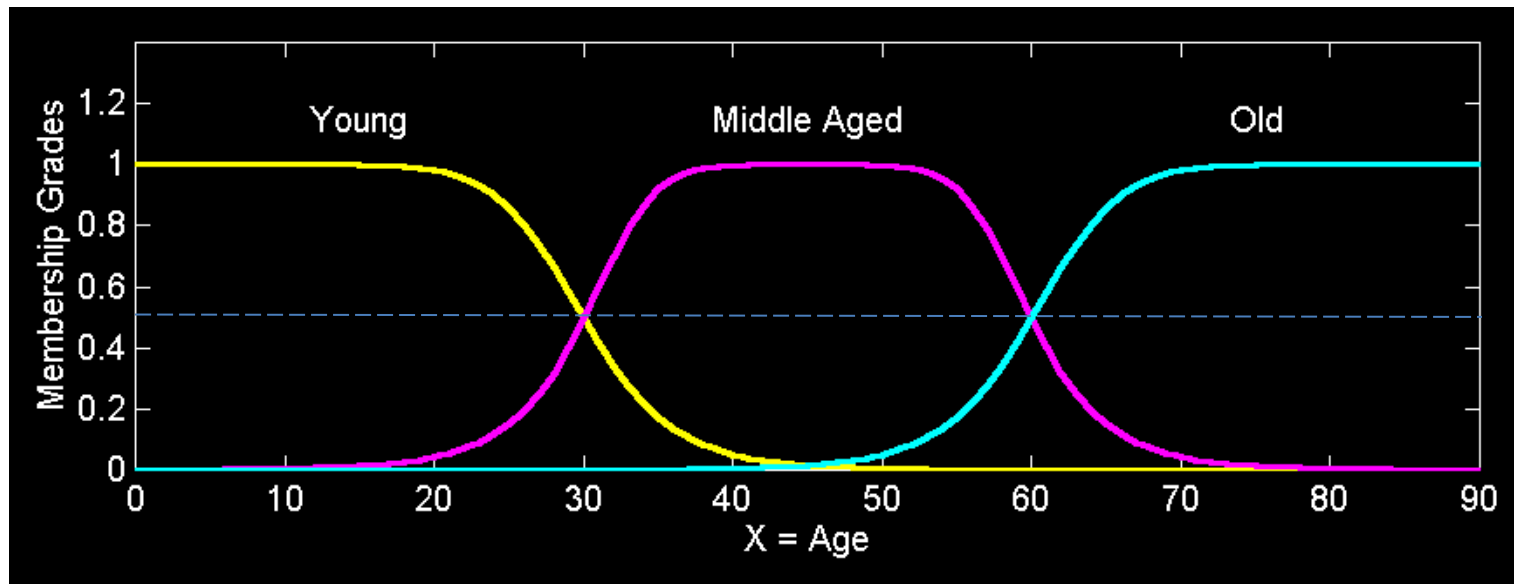
X is continuous 
$$A = \int_X \mu_A(x) / x$$

Note that Σ and integral signs stand for the union of membership grades;

“/” stands for a marker and does not imply division.

Fuzzy Partition

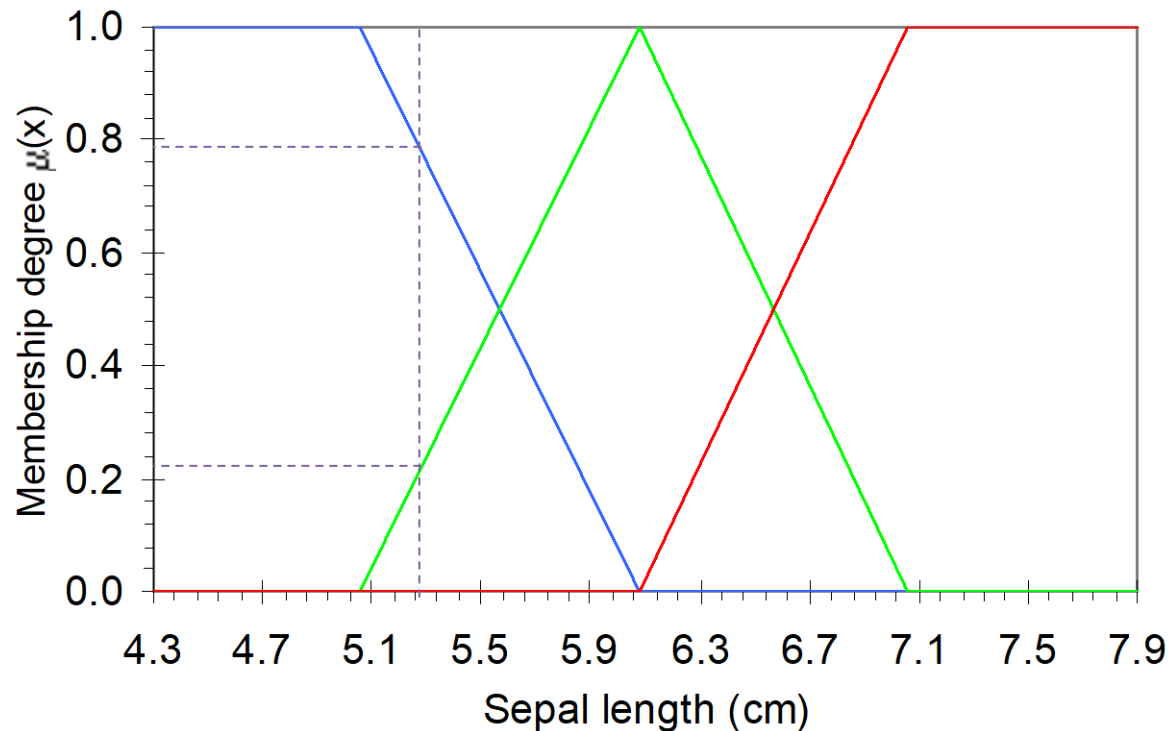
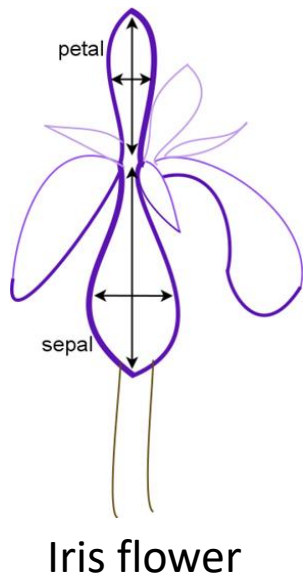
- Fuzzy partitions formed by the linguistic values “young”, “middle aged”, and “old”:



Pseudo Fuzzy Partition

A fuzzy space is a pseudo fuzzy partition when

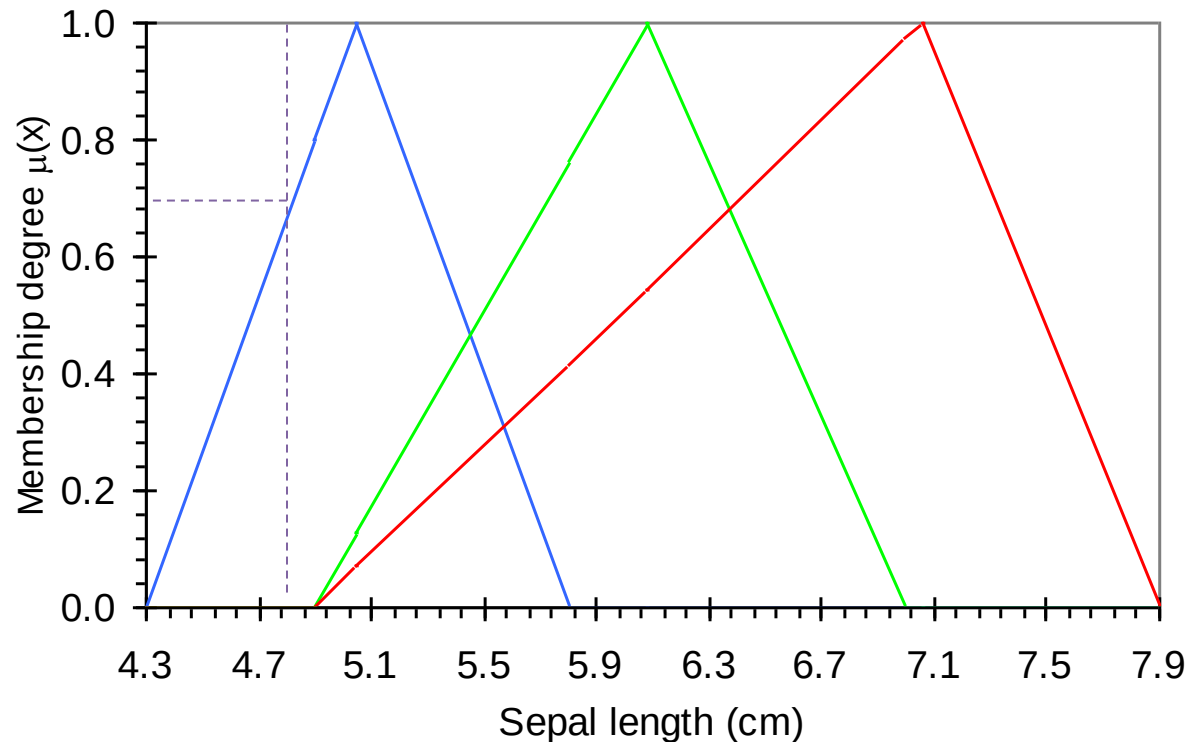
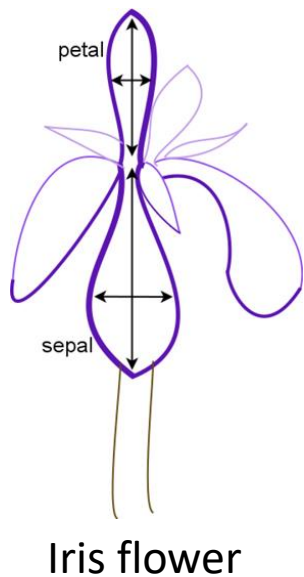
The sum of the MF values at any x is always 1



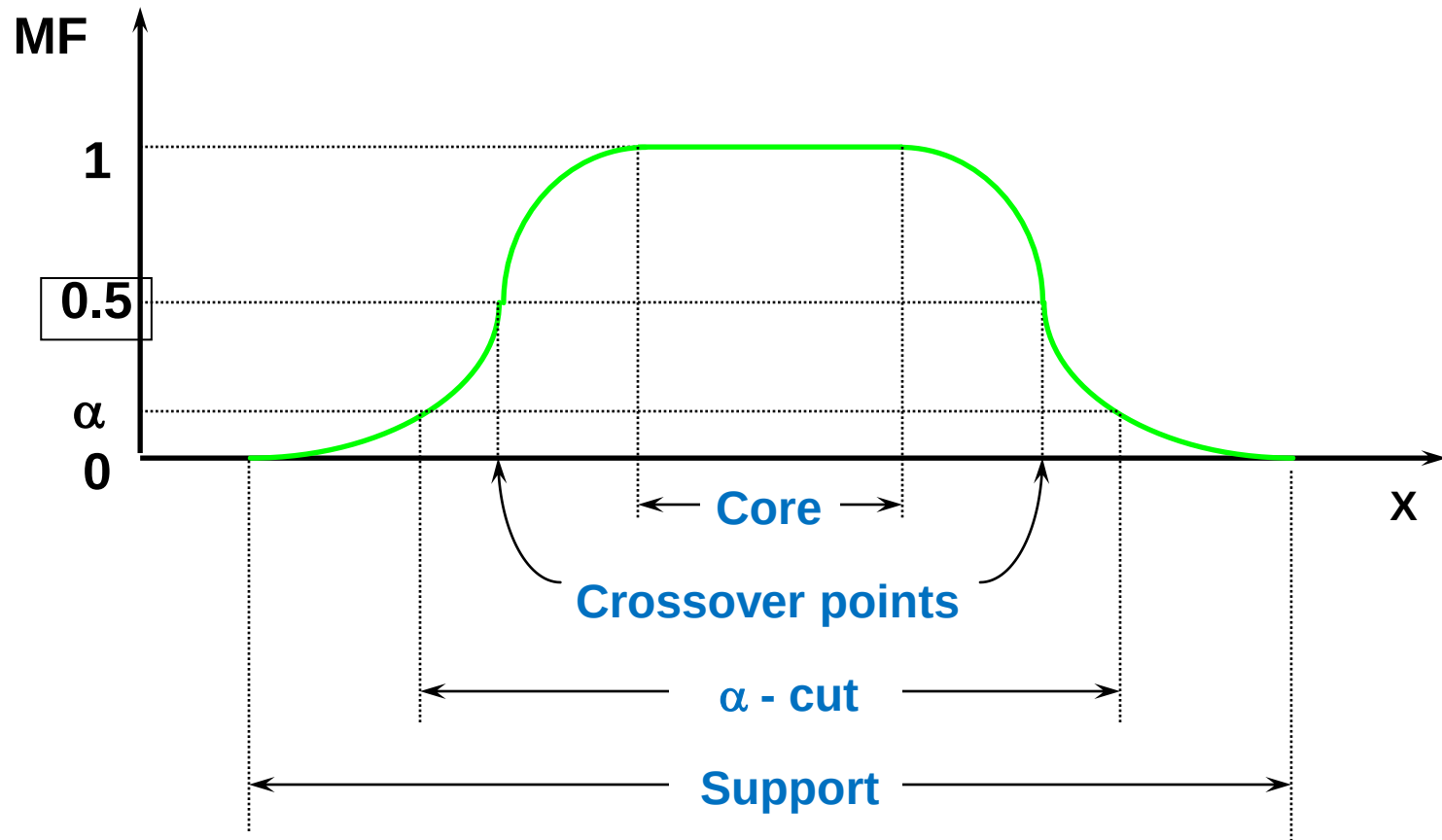
Non-Pseudo Fuzzy Partition

A fuzzy space is a non-pseudo fuzzy partition when

The sum of the MF values at any x is NOT always 1



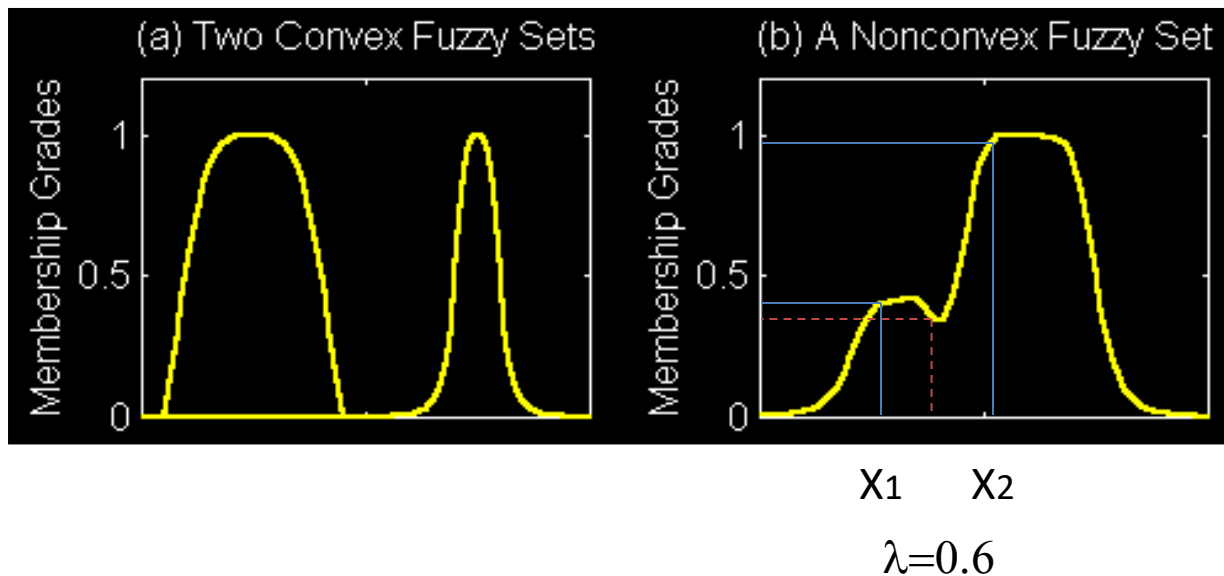
MF Terminology



Convexity of Fuzzy Sets

A fuzzy set A is convex if for any λ within $[0, 1]$:

$$\mu_A(\lambda x_1 + (1 - \lambda)x_2) \geq \min(\mu_A(x_1), \mu_A(x_2))$$



Alternatively, A is convex if all its *α -cuts* are convex.

Set-Theoretic Operations

- Subset:

$$A \subseteq B \Leftrightarrow \mu_A \leq \mu_B$$

- Complement:

$$\overline{A} = X - A \Leftrightarrow \mu_{\overline{A}}(x) = 1 - \mu_A(x)$$

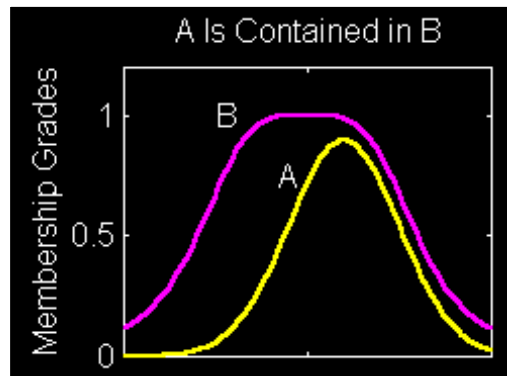
- Union: (OR, **max()**)

$$C = A \cup B \Leftrightarrow \mu_c(x) = \max(\mu_A(x), \mu_B(x)) = \mu_A(x) \vee \mu_B(x)$$

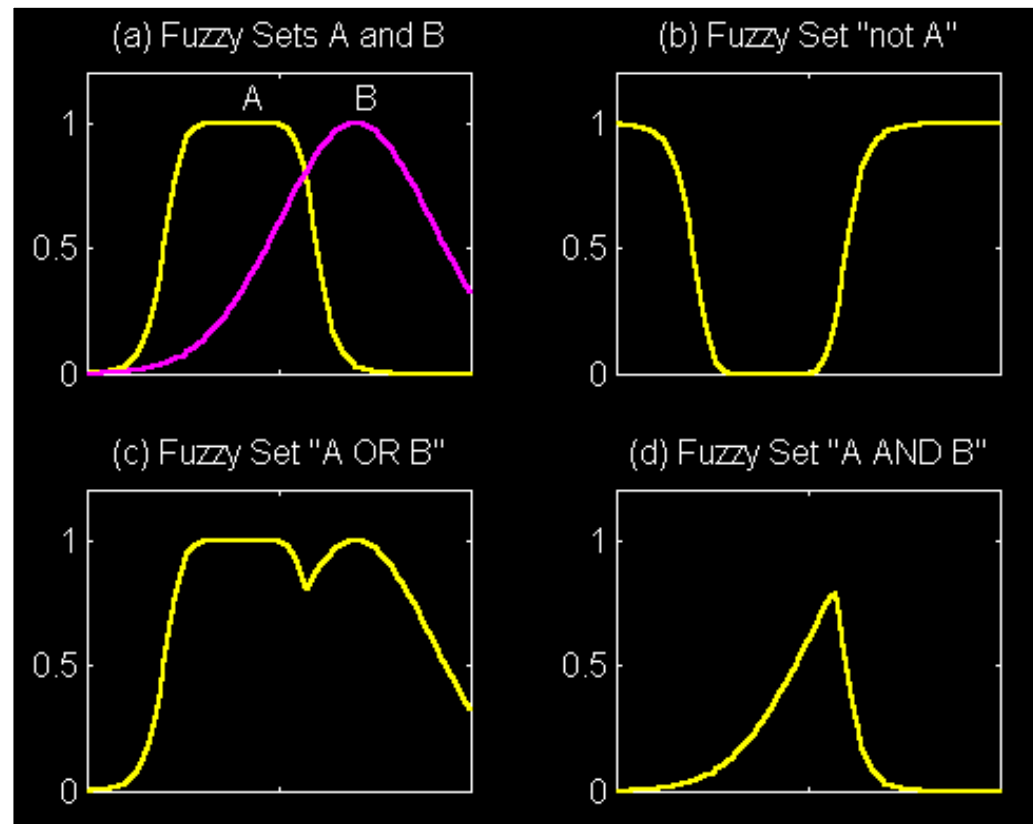
- Intersection: (AND, **min()**)

$$C = A \cap B \Leftrightarrow \mu_c(x) = \min(\mu_A(x), \mu_B(x)) = \mu_A(x) \wedge \mu_B(x)$$

Set-Theoretic Operations



$$\mu_A \leq \mu_B$$



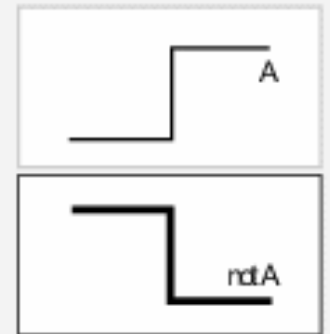
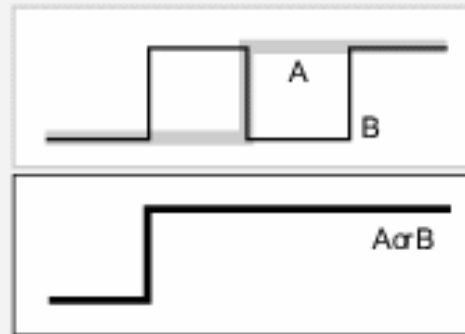
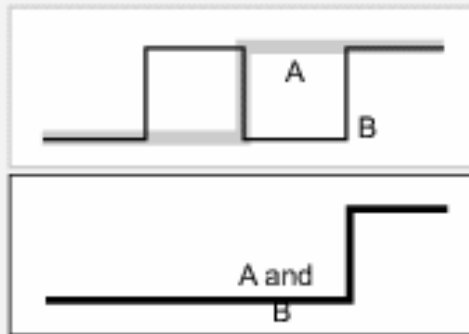
$$\max(\mu_A(x), \mu_B(x))$$

$$\min(\mu_A(x), \mu_B(x))$$

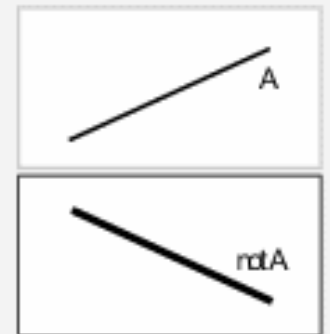
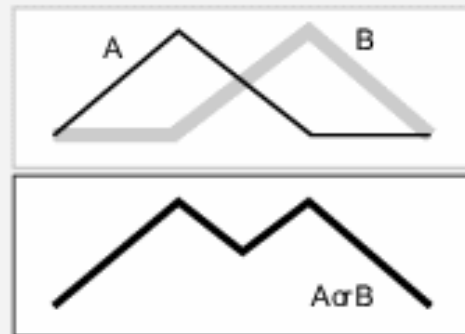
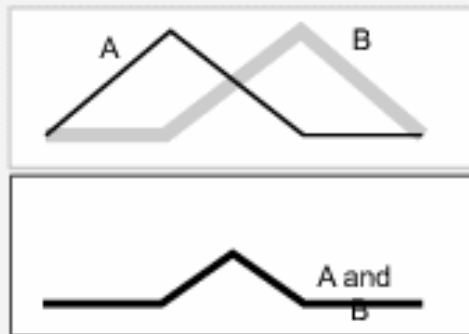
$$1 - \mu_A(x)$$

Fuzzy Logical Operation

Two-valued
logic



Multivalued
logic



AND
 $\min(A, B)$

OR
 $\max(A, B)$

NOT
 $(1 - A)$

Thank you!

